An All-Index Arithmetic Minimum Formula for OEIS A069862

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Abstract

For OEIS A069862, let $a(n)$ be the least positive k for which the decimal concatenation of $n+1,\dots,n+k$ is divisible by n. We give an exact finite arithmetic formula for this minimum for every positive integer n. In each digit interval, multiplicative-order phases reduce the coprime part to quadratic congruences, while the 2- and 5-parts reduce to a linear endpoint condition after at most three exceptional terms. Modular square roots and CRT give all admissible progressions, and ceilings select their first endpoints. Constructive prime-power lifting proves existence, infinitely many witnesses, and the bound $a(n) < 10^n - n$ for $n \geq 2$, making the formula's range explicit. The formula retains a finite minimum over arithmetic candidates; it is not an elementary expression without minimization. We also retain two infinite families of simpler exact identities. The known classification of the indices satisfying $a(n) = 2$ was recorded by David Wasserman (2005) in OEIS A069860 and is excluded from our novelty claims.

1 Definition and a modular recurrence

Let $\ell(t)$ denote the number of base-10 digits of $t \geq 1$. Define

$$C_n(0) = 0, \qquad C_n(k) = 10^{\ell(n+k)} C_n(k-1) + (n+k) \quad (k \geq 1).$$

Whenever $C_n(k) \equiv 0 \pmod{n}$, the sequence value $a(n)$ is the least such k. For $n > 1$, $C_n(1) \equiv 1 \pmod{n}$, and $a(1) = 1$.

Proposition 1 (Constant-width reduction)**.** *Put $D = \ell(n+1)$ and $L = 10^D - n - 1$. For $1 \leq k \leq L$, write $R_k = C_n(k) \bmod n$. Then all the concatenated blocks have D digits. Consequently,*

$$\begin{array}{lcll} R_k & \equiv & k & (10^D \equiv 0 \pmod{n}), \\ R_k & \equiv & k(k+1)/2 & (10^D \equiv 1 \pmod{n}), \\ R_k & \equiv & \lceil k/2 \rceil & (10^D \equiv -1 \pmod{n}). \end{array}$$

The triangular expression is an integer before reduction, even for even n.

Proof. In this range $n + i < 10^D$ and $n + i \geq n + 1 \geq 10^{D-1}$, so the recurrence modulo n is $R_k \equiv BR_{k-1} + k$ with $B = 10^D$ and $R_0 = 0$. When $B \equiv 0$, the result is k; when $B \equiv 1$, it is $\sum_{i=1}^{k} i = k(k+1)/2$. When $B \equiv -1$, induction gives $R_{2t} \equiv t$ and $R_{2t+1} \equiv t + 1$. □

2 Existence for every index

The local lifting method used for Champernowne prefixes in the repository's A075002 manuscript [1] also applies to a prefix starting at $n + 1$. The essential extra observation is that when $3 \mid n$, the removed triangular sum $n(n+1)/2$ vanishes modulo 3. We give the full argument so that no certificate from another sequence is needed.

Theorem 1 (All-index existence)**.** *For every positive integer n, there are infinitely many $t > n$ such that*

$$n \mid F_n(t), \qquad F_n(t) = (n+1)\|(n+2)\|\cdots\|t.$$

In particular, $a(n)$ exists for every $n \geq 1$.

First put $F_n(n) = 0$. In a fixed d-digit interval, let $B = 10^d$ and suppose $t+1, \dots, t+L$ all lie in that interval. Appending these blocks gives, for $L \geq 0$,

$$F_n(t+L) - F_n(t) = (B^L - 1)F_n(t) + tS_L + T_L, \tag{1}$$

where $S_L = \sum_{j=0}^{L-1} B^j$ and $T_L = \sum_{j=1}^{L} jB^{L-j}$. The telescoping identity

$$(B-1)T_L = BS_L - L \tag{2}$$

will be used only when the indicated division is legitimate.

Lemma 1 (Two local shifts)**.** *Assume the fixed-width condition in (1).*

1. *Let $p \geq 7$ be prime with $p \nmid B(B-1)$, and let $h = \operatorname{ord}_p(B)$. If $L = chp^a$, $a \geq 0$, and $p \nmid c$, then*

$$F_n(t+L) - F_n(t) \equiv -\frac{chp^a}{B-1} \pmod{p^{a+1}}.$$

   For $e \geq 1$, every multiple of $p^e \operatorname{ord}_{p^e}(B)$ preserves both $F_n(t)$ and t modulo p^e, provided the appended blocks stay in the interval.

2. *If $a = v_3(L) \geq 1$, then*

$$F_n(t+L) - F_n(t) \equiv Lt + \frac{L(L+1)}{2} \pmod{3^{a+1}}.$$

   Every multiple of $2 \cdot 3^e$ preserves both $F_n(t)$ and t modulo 3^e, under the same interval condition.

Proof. For the first assertion, the binomial theorem (or LTE) gives $B^{chp^a} \equiv 1 \pmod{p^{a+1}}$, since $B^h \equiv 1 \pmod{p}$. Thus $S_L \equiv 0$ because $B-1$ is a unit. Equations (1)–(2) give the formula. For a multiple of $p^e \operatorname{ord}_{p^e}(B)$, the same identities give $B^L = 1$, $S_L = T_L = 0$ modulo p^e and $p^e \mid L$.

For the second assertion write $B = 1 + c$ with $9 \mid c$. Expansion gives

$$S_L - L = \sum_{j \geq 1} c^j \binom{L}{j+1}, \qquad T_L - \frac{L(L+1)}{2} = \sum_{j \geq 1} c^j \binom{L+1}{j+2}.$$

The valuations of the summands are respectively at least $2j + a - v_3(j+1)$ and $2j + a - v_3((j+1)(j+2))$, both at least $a+1$. Here $v_3(j+1) \leq j$, and $v_3((j+1)(j+2)) \leq 2j-1$ for $j \geq 1$ (check $j = 1$ directly; for $j \geq 2$, $3^{2j} > (j+1)(j+2)$). Also $B^L \equiv 1 \pmod{3^{a+2}}$ by LTE. Substitute in (1). If $2 \cdot 3^e \mid L$, the displayed shift is zero modulo 3^e, and t is preserved as well. □

Proof of Theorem 1. The case $n = 1$ is immediate. Write

$$n = R\,3^e \prod_{i=1}^{s} p_i^{e_i}, \qquad R = 2^\alpha 5^\beta, \qquad 7 \leq p_1 < \cdots < p_s.$$

Let $D = \operatorname{lcm}_i \operatorname{ord}_{p_i^{e_i}}(10)$, with $D = 1$ for an empty product. Choose $d \equiv 1 \pmod{D}$ sufficiently large, so $d \geq \max(\alpha, \beta)$ and $10^{d-1} > n$. Then $B = 10^d$ is zero modulo R and is congruent to 10 modulo each $p_i^{e_i}$. All step sizes below are consequently independent of d.

Define $H_0 = 2 \cdot 3^e$ if $e > 0$, and $H_0 = 1$ otherwise; recursively put

$$H_i = \operatorname{lcm}\big(H_{i-1}, p_i^{e_i} \operatorname{ord}_{p_i^{e_i}}(10)\big).$$

For $1 \leq a < e$ let $L_a = \operatorname{lcm}(2 \cdot 3^a, R)$. For $0 \leq a < e_i$ put

$$J_{i,a} = \operatorname{lcm}\big(\operatorname{ord}_{p_i}(10) p_i^a, H_{i-1}, R\big), \qquad M_0 = R3^{\min(e,1)}.$$

An explicit displacement bound is

$$W = 2\sum_{a=1}^{e-1} L_a + \sum_{i=1}^{s}(p_i - 1) \sum_{a=0}^{e_i - 1} J_{i,a}, \qquad 9 \cdot 10^{d-1} > M_0 + W. \tag{3}$$

Empty sums are zero. Increase d within its progression to ensure (3), and let t_0 be the least multiple of M_0 at least 10^{d-1}. Every candidate below lies between t_0 and $t_0 + W < 10^d$, which proves every required fixed-width condition.

The last block gives $F_n(t_0) \equiv t_0 = 0 \pmod{R}$. If $e > 0$, then $3 \mid n, t_0$, so

$$F_n(t_0) \equiv \sum_{j=n+1}^{t_0} j = \frac{t_0(t_0+1) - n(n+1)}{2} \equiv 0 \pmod 3.$$

At a root modulo 3^a, $1 \le a < e$, choose among endpoints $t + jL_a$, $j = 0, 1, 2$. Write $L_a = 2c3^a$, where $3 \nmid c$. Lemma 1 gives

$$\frac{F_n(t + jL_a) - F_n(t)}{3^a} \equiv jc(2t+1) \pmod 3.$$

Since $3 \mid F_n(t)$ and $3 \mid n$, the triangular identity implies $t \equiv 0$ or $2 \pmod 3$. The slope $c(2t+1)$ is therefore a unit. Exactly one choice raises the root to 3^{a+1}. All endpoints remain multiples of R, preserving the last-block condition. This installs $R3^e$; the period H_0 preserves the completed 3-part.

Now process p_i in increasing order. Every prime divisor of $p_j^{e_j} \operatorname{ord}_{p_j^{e_j}}(10)$ is at most p_j, since the order divides $p_j^{e_j-1}(p_j - 1)$. Thus $p_i \nmid RH_{i-1}$. Consequently $J_{i,a} = chp_i^a$ with $h = \operatorname{ord}_{p_i}(10)$ and $p_i \nmid c$. At a root modulo p_i^a (vacuous for $a = 0$), Lemma 1 shows that the endpoints $t + jJ_{i,a}$, $0 \le j < p_i$, sweep all next p_i-adic digits, with unit normalized slope $-ch/(B-1)$ modulo p_i. Choose the unique root modulo p_i^{a+1}. The multiples of H_{i-1} preserve all earlier odd prime powers, and those of R preserve the last-block condition. After installing $p_i^{e_i}$, the period H_i preserves it too. Induction yields $n \mid F_n(t)$.

We have $n < t < 10^d$, hence the explicit bound $a(n) \le t - n < 10^d - n$. There are arbitrarily large admissible d; their disjoint digit intervals give infinitely many endpoints. This proves the theorem. □

Corollary 1 (A uniform bound)**.** *For every $n \ge 2$,*

$$1 \le a(n) < 10^n - n.$$

Proof. Retain the notation in the proof, and put $K = \operatorname{lcm}(R, H_s)$. Every L_a and $J_{i,a}$ divides K, and

$$K \le 2n \prod_i \varphi(p_i^{e_i}) \le 2n^2.$$

The elementary inequalities $p^f \ge 1 + f(p-1)$ and $\prod_j (1 + x_j) \ge 1 + \sum_j x_j$ for nonnegative x_j give

$$2\max(e-1, 0) + \sum_i (p_i - 1)e_i \le n - 1.$$

Thus $W \le 2n^2(n-1)$ and $M_0 \le n$. Let $b = \lceil \log_2 n \rceil$ and choose

$$d = 1 + D\lceil b/D \rceil.$$

Then $d \ge b + 1$, so $10^{d-1} \ge 10^b \ge n^3 > n$ and $9 \cdot 10^{d-1} > n + 2n^2(n-1)$. All conditions in the construction hold. Also $D \le \prod_i \varphi(p_i^{e_i}) \le n - 1$ (including the empty-product case). If $D \ge b$, then $d = D + 1 \le n$. If $D < b$, then $d \le b + D \le 2b - 1 \le n$; the last inequality holds for $n \ge 3$, and $n = 2$ falls in the first case. The constructed endpoint therefore satisfies $n < t < 10^d \le 10^n$. □

Proposition 2 (Exact minimum for every index)**.** *Every value $a(n)$ can be determined in at most n^3 modular single-block transitions, together with at most n binary block jumps.*

Proof. The case $n = 1$ is immediate. For $n \ge 2$, process digit intervals in increasing order, starting with the integer $n + 1$ and stopping before 10^n. Within a d-digit interval, the state $(F_n(t) \bmod n, t \bmod n)$ lies in a set of size n^2 and evolves by

$$(x, y) \longmapsto (10^d x + y + 1,\ y + 1) \pmod n.$$

Test the residue after each actual appended block. If it is zero, the first such endpoint gives the exact minimum. If a state repeats before a zero occurs, determinism implies that the remaining orbit in this interval has no zero. Jump over the unused part of the interval by binary exponentiation of its affine transition and proceed to the next digit width. A repeated state occurs within n^2 transitions, irrespective of whether 10^d is invertible modulo n. There are at most n digit intervals, and Corollary 1 guarantees a zero before they end. No guessed numerical search limit or assumed general minimum formula is used. □

3 An all-index arithmetic formula for the minimum

We now determine the minimum itself for every index. The expression below uses multiplicative orders, modular square roots, CRT, ceilings, and a finite minimum over explicitly constructed residue classes. It is a *finite arithmetic formula*, not an elementary expression without finite minimization. It does not enumerate concatenation lengths, and no unknown earlier sequence values occur in it.

3.1 Quantities depending only on the index

Fix $n \geq 2$ and write

$$n = Rm, \qquad R = 2^{\alpha}5^{\beta}, \qquad \gcd(m, 10) = 1, \qquad D = \ell(n+1).$$

For each $D \leq d \leq n$, put

$$B_d = 10^d, \qquad t_d = \max(n, 10^{d-1} - 1), \qquad L_d = 10^d - 1 - t_d.$$

The actual d-digit endpoints are $t_d + 1, \ldots, t_d + L_d$. For any $B > 1$ and integer $r \geq 0$, define the integers

$$S_B(r) = \frac{B^r - 1}{B - 1}, \qquad T_B(r) = \frac{B^{r+1} - (r+1)B + r}{(B-1)^2}. \tag{4}$$

In particular $S_B(0) = T_B(0) = 0$. An explicit, nonrecursive expression for the prefix preceding this digit interval is

$$c_d = \sum_{e=D}^{d-1} \big(t_e S_{B_e}(L_e) + T_{B_e}(L_e)\big) 10^{\sum_{f=e+1}^{d-1} f L_f}. \tag{5}$$

The empty sum is zero. This is simply the sum of the complete fixed-width words, shifted by the number of digits following them. Thus $c_d = F_n(t_d)$. Computation may replace c_d by its residue modulo n.

For the remainder of this subsection fix d and abbreviate $B = B_d, t = t_d, c = c_d$. Put

$$E = \max\left(1, \lceil \alpha/d \rceil, \lceil \beta/d \rceil\right), \quad h = \operatorname{ord}_m(B), \quad s = S_B(h), \quad v = T_B(h),$$

where $h = 1$ when $m = 1$. Equivalently, for $m > 1$, $h = \operatorname{ord}_m(10)/\gcd(\operatorname{ord}_m(10), d)$. For $0 \leq j < h$ define

$$V_j = B^j c + t S_B(j) + T_B(j),$$

and the quadratic

$$Q_j(z) = hsz^2 + \big(2(t+j)s + 2v - hs\big)z + 2V_j. \tag{6}$$

Let $\mathcal{R}_j$ be a set of disjoint residue classes (r, q), with $q \mid m$, whose union is precisely the roots of $Q_j(z) \equiv 0 \pmod m$. The explicit square-root construction below gives at most $2^{\omega(m)}$ such classes; for $m = 1$ use $\mathcal{R}_j = \{(0, 1)\}$.

Write $[x]_u$ for the least nonnegative residue modulo u, with $[x]_1 = 0$ and all modular-inverse expressions modulo 1 interpreted as 0. Define

$$\rho = [-B(B-1)^{-1}]_R, \qquad g = \gcd(h, R), \qquad u = R/g.$$

Discard a phase j if $g \nmid \rho - t - j$. For every other phase put

$$b_j = \left[\frac{\rho - t - j}{g}(h/g)^{-1}\right]_u.$$

For each $(r, q) \in \mathcal{R}_j$, the CRT combination and the first eligible endpoint are the explicit expressions

$$w = r + q[(b_j - r)q^{-1}]_u, \qquad P = hqu, \tag{7}$$

$$x_0 = t + j + hw, \qquad X_{d,j,r,q} = x_0 + P\left\lceil \frac{t + E - x_0}{P} \right\rceil. \tag{8}$$

All inverses here exist in the stated moduli. Define the finite sets

$$\mathcal{E}_d = \{t + r : 1 \leq r < E,\ r \leq L_d,\ n \mid B^r c + t S_B(r) + T_B(r)\},$$
$$\mathcal{C}_d = \{X_{d,j,r,q} : 0 \leq j < h,\ (r, q) \in \mathcal{R}_j,\ g \mid \rho - t - j,\ X_{d,j,r,q} \leq 10^d - 1\}.$$

Theorem 2 (All-index minimum formula)**.** *For $n \geq 2$, the following finite set is nonempty, and*

$$\boxed{a(n) = \min \bigcup_{d=D}^{n} (\mathcal{E}_d \cup \mathcal{C}_d) - n.} \tag{9}$$

Together with $a(1) = 1$, this determines the entire sequence. For each digit width there are at most three exceptional endpoints and at most $h\,2^{\omega(m)}$ progression candidates. All quantities in the formula are specified by n, with no test of successive lengths up to $a(n)$.

3.2 Proof of completeness and minimality

For a fixed width, write $G(r) = B^r c + tS_B(r) + T_B(r)$, initially for nonnegative integers r. This is the fixed-width continuation of the prefix; it equals the actual concatenation whenever $r \leq L_d$. Multiplying by $(B-1)^2$ gives

$$(B-1)^2 G(r) = \big((B-1)^2 c + t(B-1) + B\big)B^r - (t+r)(B-1) - B. \tag{10}$$

Since $B-1$ is a unit modulo R and $R \mid B^r$ for $r \geq E$,

$$R \mid G(r) \quad \Longleftrightarrow \quad t + r \equiv \rho \pmod{R} \qquad (r \geq E). \tag{11}$$

The only omitted positive offsets are $1, \ldots, E-1$, exactly those tested in $\mathcal{E}_d$. Because $R \leq n < 10^d$, we have $\alpha < 4d$ and $\beta < 2d$, so $E \leq 4$.

Now work modulo the odd number m. By definition $B^h \equiv 1$. Appending h fixed-width blocks maps the state (C, y) to $(C + ys + v, y + h)$ modulo m. Starting after j blocks and iterating this affine map z times gives

$$G(j + hz) \equiv V_j + z((t+j)s + v) + hs\frac{z(z-1)}{2} \pmod{m}.$$

Multiplication by 2 is invertible modulo m, so

$$m \mid G(j + hz) \quad \Longleftrightarrow \quad Q_j(z) \equiv 0 \pmod{m}. \tag{12}$$

Every nonnegative offset has a unique expression $j + hz$ with $0 \leq j < h$ and $z \geq 0$. Equation (11) becomes $hz \equiv \rho - t - j \pmod{R}$, which is soluble exactly when the stated divisibility by g holds; in that case it is $z \equiv b_j \pmod{u}$. As $q \mid m$ and $u \mid R$, their moduli are coprime. Equation (7) therefore gives exactly all compatible solutions $z \equiv w \pmod{qu}$, hence endpoints $x \equiv x_0 \pmod{P}$.

The ceiling in (8) selects the least such endpoint at least $t + E$. If it exceeds $10^d - 1$, that progression has no endpoint in the actual digit interval. Otherwise it is the smallest endpoint of its progression in the interval. Any selected endpoint has $x - t \geq 1$; since $x - t \equiv j \pmod{h}$ and $j < h$, its corresponding z is nonnegative. Thus the use of the iterated-map identity is legitimate.

These implications prove both that every retained candidate is divisible by n and that every divisible endpoint in this digit interval is either in $\mathcal{E}_d$ or no smaller than a candidate in $\mathcal{C}_d$. Corollary 1 ensures at least one such endpoint before 10^n. Taking the minimum over the stated widths proves (9), including exclusion of *all* shorter lengths, not only those of the same digit width.

3.3 Explicit quadratic classes, including singular roots

For completeness, $\mathcal{R}_j$ need not be treated as an unspecified congruence-solving operation. The following gives it by modular square roots and CRT for any quadratic $Az^2 + Bz + C$ modulo an odd m. Factor $m = \prod p^e$ and handle each prime power separately. First divide A, B, C and p^e by their largest common power of p. The resulting congruence is equivalent as a condition on the integer z; its reduced modulus suffices to describe the original solution set. If the reduced modulus is 1, the sole class is $(0, 1)$. Otherwise write it as p^f, so at least one coefficient is a unit.

1. If $p \mid A, B$, then $p \nmid C$ and there are no roots.
2. If $p \mid A$ and $p \nmid B$, let y be the unique root of $y^2 \equiv B^2 - 4AC \pmod{p^f}$ satisfying $y \equiv B \pmod{p}$. The sole class is
$$z \equiv -2C(B + y)^{-1} \pmod{p^f}.$$
The indicated square root exists and is unique by simple-root lifting. The denominator is a unit. Substitution verifies the formula; conversely a root z gives $y = 2Az + B$ and $(B + y)z = -2C$, proving completeness.

3. If $p \nmid A$, put $\delta = [B^2 - 4AC]_{p^f}$. For $\delta = 0$, there is one class

$$z \equiv -B(2A)^{-1} \pmod{p^{\lceil f/2 \rceil}}.$$

For $\delta \neq 0$ of odd p-adic valuation there are no roots. Otherwise put $v_p(\delta) = 2s < f$. If the unit δ/p^{2s} is not a square modulo p, there are no roots. If it is, take either unit square root y modulo p^{f-2s}. The two classes are

$$z \equiv (-B \pm p^s y)(2A)^{-1} \pmod{p^{f-s}}.$$

These cases follow from the equivalent square congruence $(2Az + B)^2 \equiv \delta \pmod{p^f}$.

Every prime contributes at most two classes, with moduli powers of that prime. Combining them by CRT gives disjoint classes (r, q) with $q \mid m$ and at most $2^{\omega(m)}$ possibilities. This includes all free higher digits of singular roots without enumerating them.

3.4 An example requiring several digit intervals

For $n = 451 = 11 \cdot 41$, the formula gives the following complete stable classes of endpoints. Here $R = 1$, so there are no transient exceptions.

d	$c_d \bmod 451$	endpoint residue classes
3	0	$451, 1902, 1133, 3794, 4345, 1066, 1287, 1748, 3179, 450 \pmod{4510}$
4	241	$\varnothing$
5	196	$\varnothing$
6	63	$2084, 555, 1501, 137, 478 \pmod{2255}$

None of the width-3 classes meets $[452, 999]$. The width-4 and width-5 quadratics have no compatible roots at all. At width 6, the ceiling formula gives first endpoints $101304, 102030, 100721, 101612, 101953$, respectively. Hence the least endpoint is 100721 and $a(451) = 100721 - 451 = 100270$. The formula explains the long gap by whole empty digit intervals, without testing their individual lengths.

4 First exact family: a last-block principle

Theorem 3 (Decimal-smooth moduli)**.** *For any $n \geq 1$ with $D = \ell(n+1)$, if $n \mid 10^D$, then*

$$\boxed{a(n) = n}.$$

In particular $a(10^r) = 10^r$ for all integers $r \geq 0$.

Proof. Every $n + k$ for $k \geq 1$ has at least D digits, so $n \mid 10^{\ell(n+k)}$. The recurrence thus yields $C_n(k) \equiv k \pmod{n}$ *for all* $k \geq 1$, including after any digit-width transitions. All $1 \leq k < n$ are nonzero modulo n, while $k = n$ is zero. □

Corollary 2. *The formula holds whenever $n = 2^u 5^v$ and $\max(u, v) \leq \ell(n+1)$. Examples are $a(20) = 20$, $a(25) = 25$, $a(125) = 125$, and $a(1000) = 1000$.*

5 Second exact family: an alternating residue

Theorem 4 (An infinite family with $a(n) = 2n - 1$)**.** *For every integer $j \geq 0$ let $D = 6j + 3$ and $n = (10^D + 1)/7$. Then n is a positive integer and*

$$\boxed{a\left(\frac{10^{6j+3} + 1}{7}\right) = \frac{2(10^{6j+3} + 1)}{7} - 1}.$$

In particular, $a(143) = 285$.

Proof. Since $10 \equiv 3 \pmod 7$, $3^3 \equiv -1 \pmod 7$, and 3 has order 6 modulo 7, we have $7 \mid 10^D + 1$. For $D \geq 3$ and $n = (10^D + 1)/7$, the inequalities

$$10^{D-1} \leq n + 1 \quad \text{and} \quad 3n - 1 < 10^D$$

hold. Hence all terms $n + 1, \ldots, n + (2n - 1)$ have exactly D digits. Moreover $10^D = 7n - 1 \equiv -1 \pmod n$. Proposition 1 now yields $C_n(k) \equiv \lceil k/2 \rceil \pmod n$ for every $1 \leq k \leq 2n - 1$. The first value in this range whose ceiling is divisible by n is precisely $k = 2n - 1$. Thus no shorter prefix is divisible by n, and that prefix is. □

6 A companion exact criterion

Proposition 3 (Odd prime powers)**.** *Suppose $n > 1$ is an odd prime power, $D = \ell(n+1)$, $n \mid 10^D - 1$, and $10^D \geq 2n$. Then*

$$\boxed{a(n) = n - 1}.$$

For example $a(9) = 8$ and $a(11) = 10$.

Proof. The hypotheses ensure $L = 10^D - n - 1 \geq n - 1$. The constant-width recurrence gives $C_n(k) \equiv k(k+1)/2 \pmod{n}$ for $k \leq n - 1$. Since $n = p^e$ with p odd, n divides $k(k+1)/2$ if and only if p^e divides k or $k+1$, because these consecutive factors are coprime. The first such positive k is $n - 1$. □

7 Prior art, verification, and limitations

The classification of *all* indices with $a(n) = 2$ follows from the two-block congruence $C_n(2) \equiv 10^{\ell(n+2)} + 2 \pmod{n}$. The equivalent complete formula was published as a formula for OEIS A069860 by David Wasserman in 2005. We do not present it as a new theorem or claim credit for that classification.

The dependency-free script `minimum_formula.py` evaluates Theorem 2 and exposes the residue classes with `-explain`. Its output agrees with all 10,000 terms in the OEIS b-file. Its quadratic solver is checked exhaustively modulo every odd integer at most 31, with additional singular and higher-prime-power controls. Arbitrary-state checks test both directions of the full residue-class classification. The separate `least.py` state-orbit method and direct enumeration provide independent minimum checks. The script `verify_existence.py` independently recomputes constructive witnesses, and `verify.py` checks the older families. Finite checks supplement the proofs. The all-index result is the finite arithmetic formula (9); a simpler elementary expression without finite minimization is not established. Broader literature priority and external peer-review acceptance remain unverified.

References

[1] Q. Wang, *Divisibility of Concatenated Arithmetic Progressions and Champernowne Prefixes*, September 11, 2026, repository manuscript, https://github.com/dongxuelian2/oeis-papers/tree/main/papers/A075000-A075002.

[2] OEIS Foundation Inc., *A069862: Smallest k such that n divides the concatenation of numbers from $(n+1)$ to $(n+k)$*, https://oeis.org/A069862 (accessed October 22, 2026).

[3] D. Wasserman, formula for *A069860: Numbers m that divide the concatenation of $m+1$ and $m+2$*, OEIS entry, August 25, 2005, https://oeis.org/A069860.